Symantyx gateway/ pihole and wireguard

Saturday, October 10, 2026 2:00 AM

1. Symantyx EssenceContainer for the gateway

yaml
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```yaml
EssenceContainer:
  id: "gateway.wireguard.pihole.v1"
  domain: "network_gateway"
  description: "Encrypted remote-access gateway with Pi-hole DNS and WireGuard VPN."
program:
  name: "SymantyxGatewayNode"
  version: "1.0"
  executable: "/usr/bin/wg-quick"
action:
  type: "start_gateway"
  parameters:
    vpn_interface: "wg0"
    vpn_listen_port: 51820
    vpn_address: "10.8.0.1/24"
    lan_interface: "wlan0"
    lan_ip: "192.168.1.131"
    dns_service: "pihole-FTL"
    dns_ip: "192.168.1.131"
    semantic_tags:
      - "vpn"
      - "dns"
      - "gateway"
      - "remote_access"
target:
  object_type: "GatewayNode"
  locator:
    host: "raspberrypi"
    ip: "192.168.1.131"
path:
  invocation_sequence:
    - "systemctl start wg-quick@wg0"
    - "systemctl start pihole-FTL.service"
  reproducibility_notes: "WireGuard + Pi-hole must be installed and configured before auto-start."
flags:
  display_profile: "gateway"
  zoom_level: "operator"
  mode: "execution"
  tool_settings: {}
timers:
  capture_timestamp: null
  session_id: null
outputs:
  vpn_status: null
  dns_status: null
  gateway_reachability: null
evidentiary_notes:
  repeatability: "Same config yields same encrypted gateway behavior."
  limitations: "Router port forwarding and WAN IP must be correct."
  admissibility_support: "WireGuard and Pi-hole logs provide forensic trace."
```

2. Gateway Operator Manifest (high-level)

yaml
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```yaml
GatewayOperatorManifest:
  name: "HomeRemoteAccessGateway"
  version: "1.0"
  components:
    - id: "wireguard_server"
      type: "vpn_server"
      interface: "wg0"
      listen_port: 51820
      address: "10.8.0.1/24"
      nat_backend: "nftables"
      lan_interface: "wlan0"
    - id: "pihole_dns"
      type: "dns_filter"
      service: "pihole-FTL"
      ip: "192.168.1.131"
      ui: "http://192.168.1.131/admin"
    - id: "router_binding"
      type: "port_forward"
      router_model: "Linksys"
      wan_endpoint: "64.32.35.51:51820"
      forward_rule:
        protocol: "udp"
        external_port: 51820
        internal_ip: "192.168.1.131"
        internal_port: 51820
  behaviors:
    - name: "AutoStartOnBoot"
      sequence:
        - "systemctl enable wg-quick@wg0"
        - "systemctl enable pihole-FTL.service"
    - name: "RemoteAccess"
      description: "VPN clients reach LAN via WireGuard and use Pi-hole DNS."
```

3. Shim wrapper that auto-starts WireGuard + Pi-hole

Use a simple systemd unit as your “shim”:
ini
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```ini
# /etc/systemd/system/symantyx-gateway.service
[Unit]
Description=Symantyx Gateway Shim (WireGuard + Pi-hole)
After=network-online.target
[Service]
Type=oneshot
ExecStart=/usr/bin/wg-quick up wg0
ExecStart=/bin/systemctl start pihole-FTL.service
RemainAfterExit=yes
[Install]
WantedBy=multi-user.target
```

Enable it:
bash
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```bash
sudo systemctl enable symantyx-gateway.service
```

```
sudo systemctl start symantyx-gateway.service
```

This becomes your “gateway shim”—one operator that brings up VPN + DNS together.

4. Integrate Pi-hole with your VPN clients

On the Pi (server):

Your wg0.conf already has:
ini
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```
[Interface]
Address = 10.8.0.1/24
```

On the Windows client:

Set DNS to Pi-hole via the tunnel:
ini
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```
[Interface]
PrivateKey = WLGzzj8H0FRNqNlLsdl2jQ+clewhKhcWUZzhtyzleHc=
Address = 10.8.0.2/32
DNS = 10.8.0.1   # Pi-hole via VPN

[Peer]
PublicKey = +Z6dIuSipUf0GVlvzWAFQATE7jsY4wnMFw0qWC2/ZSY=
Endpoint = 64.32.35.51:51820
AllowedIPs = 0.0.0.0/0
PersistentKeepalive = 25
```

When the tunnel is active:

- All your client’s traffic goes through WireGuard
- All DNS queries go to Pi-hole at 10.8.0.1
- Pi-hole filters ads/tracking for that VPN client

Everyone else at home keeps using normal internet unless you point their DNS to the Pi.

If you want, we can next:

- Add a second EssenceContainer for “ClientNode”
- Encode the gateway as a Hypertuple for your crypto-geometry docs
- Or build a tiny sym.ps1-style shim on Windows that toggles the tunnel + RDP in one shot.

From <https://copilot.microsoft.com/projects/hEPscG4ByJbVvs5xxJiL5/chats/Yt8ianMUpZG73ws3suNCP>